

Negotiating Open Science: The Open Science Framework as a Technology-in-Practice

Johanna Cohoon
jlcohoon@utexas.edu
University of Texas at Austin
USA

ABSTRACT

The open science movement promotes use of digital technology to increase the efficiency, inclusivity, and quality of scientific research. Developers of these platforms often advocate for open science on the grounds that it is in keeping with scientific values, specifically referencing Mertonian norms. However, many scientists are agnostic toward open science; as policies and technology enforce the movement's aims of sharing openly, they seek to protect the research they view as their own. My dissertation work studies the enactment of open science by a variety of stakeholders in the Open Science Framework (OSF)—its developers, its users, and also its non-users. Through interviews, trace data collection, and observation of these various populations, I will explore how these stakeholders construct different technologies-in-practice. By taking the OSF not as a given, but as a technology whose purpose and effects are structured by the constraints and resources of its stakeholders, I will unpack the tensions among open science advocates, agnostics, and their use of technology. With these insights, science policy and open systems can be designed to better accommodate the diverse concerns of stakeholders.

CCS CONCEPTS

• **Software and its engineering** → **Collaboration in software development**; • **Information systems** → **Collaborative and social computing systems and tools**; • **Social and professional topics** → **User characteristics**.

KEYWORDS

Open science, Technologies-in-practice, Science of science

ACM Reference Format:

Johanna Cohoon. 2021. Negotiating Open Science: The Open Science Framework as a Technology-in-Practice. In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '21 Companion)*, October 23–27, 2021, Virtual Event, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3462204.3481785>

1 INTRODUCTION

Advocates for open science call attention to the technical advances of our time while noting that they have not been harnessed for transparency and democracy in research—values open science advocates

espouse as central to the scientific ethos [16]. These advocates introduce policies and infrastructure to engender a more open science, citing a reverse salient on the front of scientific achievement [10]. In response, a host of policies, initiatives, and infrastructure efforts like Plan S and NSF's data sharing policies have arisen over recent decades to promote open science because it “increases transparency and reliability, facilitates more effective collaboration, accelerates the pace of discovery, and fosters broader and more equitable access to scientific knowledge and to the research process itself” [15, p. 12].

As the open science movement gains popularity, studies have also shown that some researchers have grievances with this rising tide. Differences in values between open science advocates and agnostics might explain these protests [11]. For instance, data show that scientists often especially value their reputation and career advancement. Consistent with findings from academic open source work [9], Recker, Müller, Trixa, and Schumann emphasize in their literature review “the importance of a reward system making it possible for researchers to gain recognition for sharing their data” [18, 4]. Furthermore, while open science advocates frequently reference adherence to open science norms as “just science” [8, 2], science and technology scholars indicate that those norms are more akin to an ideology than a true representation of institutionalized scientific behaviors [14].

While some proponents claim that open science must be maintained as a new *modus operandi*, the threads of ideology and technological determinism in open science advocacy beg investigation. MacKenzie and Wajcman noted that the presence of a reverse salient indicates positionality—the concept “makes sense only if a technological system is seen as oriented to a goal” [12, 18]. But, blanket policies like those from funders and publishers that push these technologies assume a common goal among stakeholders that may not exist. “E-science is oriented toward particular cultures,” argue Beaulieu, Scharnhorst, and Wouters, and it is liable to put researchers through a “one-size-fits-all steamroller” [2, 676]. Importantly, open science extends the boundaries of science, inviting newcomers like citizen scientists to the research process—this further expands the stakeholders that may use and contribute to open technologies, recasting the tools not just as academic infrastructure but possibly as civic tech as well [3].

Given these tensions, I am conducting a qualitative study of the enactment of open science, exploring how different stakeholders in one open science platform, the Open Science Framework (OSF) approach the movement and the technology. By attending to both open science advocates' and agnostics' perspectives, this work will shed light on what open science means in practice—not

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW '21 Companion, October 23–27, 2021, Virtual Event, USA

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8479-7/21/10.

<https://doi.org/10.1145/3462204.3481785>

just in its idealized form. Through this work, I anticipate deriving implications for design and policy making.

2 THEORETICAL FRAMEWORK AND RESEARCH QUESTIONS

In approaching this work, I adopt a theoretical framework built on structuration theory [5] and technologies-in-practice [17]. In brief, my theoretical framework is one which encourages the researcher to consider how human agents draw on their material and social contexts to affect change.

Structuration theory proposes that structure is recursive and dual in nature [5]. Giddens argues that human agents act on the world and, in doing so, create the world that they will act within next. He defines agents' cyclical, perpetual use and subsequent (re)production of rules and resources (i.e. structures) as the duality of structure. To emphasize the fact that a structurational model of technology is concerned with process—how structures constrain and enable agents as those agents create and recreate those structures—Orlikowski employs the term *technology-in-practice* [17]. A *technology-in-practice* is one of the many structures an agent might enact by interacting with a technology; it is established through routine and repeated interactions with a technology.

Studying the publisher's role in modern scholarly communication, Stewart, Procter, Williams, and Poschen state, "Web 2.0 is not a given: it is driven by people creating, investing in and experimenting with ideas" [20, 416]. The study proposed here assumes a similar perspective: open science behaviors are not a given, they are yielded through the continued interaction of people, technology, and their research. The social and technical elements of science are viewed in the present research as structural influences that shape and are shaped by one another. Crucially, because of the role of policy in the case of open science, non-users of open technology must also be considered when understanding the enactment of open science. "Researchers are not generally passive users of technology but are more likely to shape, reshape, design, redesign, and repurpose technologies, even if they are also shaped by them," write Meyer and Schroeder [13, 36].

To study this mutual shaping, I am conducting a qualitative exploration of how developers and potential users of open science technology negotiate between their own varied interests and the constraints or affordances of open technology. Situating this negotiation within my research "site," the OSF, and translating it into structurational terms, I ask three research questions:

RQ1: What structures do developers, users, and non-users draw on when they interact with the OSF?

RQ2: How do developers, users, and non-users of the OSF enact different technologies-in-practice?

RQ3: What value tensions emerge (if any) among OSF stakeholders and how do they respond (or not respond) to those tensions?

3 THE OPEN SCIENCE FRAMEWORK

The OSF (osf.io) is a web platform developed by the Center for Open Science (COS). Through the OSF and their many activities, COS candidly attempts to change science practice by engaging with researchers themselves. "Cultural revolutions are more likely to persist if they emerge from the community rather than being

forced on to the community," wrote the scientist co-founders of COS [1, 1]. However, the plans and actions of COS demonstrate that the open science revolution is not emerging unprompted from individual researchers—the non-profit directly engages with scientists and prompts them via the OSF's technical features to behave openly. Importantly, COS's cofounders, Spies and Nosek, are vocal advocates for open science and are among those who frequently reference Mertonian norms as scientific ideals. Furthermore, COS employees espouse a view of open science as "just science" [8, 2].

The OSF was developed to be "an infrastructure for collaborating, documenting, archiving, sharing, and registering scientific materials" [19, 23]. Users of the OSF can create projects to organize their research. Projects are made up of components that have most of the same technical features as their parent projects: a title, a contributor list, a wiki, file storage, tags, a unique URL, and timestamped activity logs. Users can collaborate with one another on projects and components of those projects. Spies designed the platform so that users can render projects or their components private from non-collaborators by changing the default setting of a project/component.

The OSF includes a registration feature. Registrations are time-stamped versions of projects that are immutable by any user. Because they cannot be changed, registrations are used to show the status of a project at a given point in time—they are useful to show at the conclusion of a project that all a priori hypotheses have been tested as planned.

The OSF is available online for individual users to sign up. Additionally, the platform can also be leveraged by organizations in several ways. Conferences and meetings can use OSF Meetings to host talks, posters, and presentations. Preprint service providers can use OSF Preprints to host their content. Journals and societies can also use the platform to collect and host registered studies via OSF Registries. Institutions can coalesce and disseminate their members' research via OSF Institutions.

4 METHODS

In summary, the research will involve observation and interviews of developers, users, and also non-users of the Open Science Framework as well as analysis of trace data. Results will be analyzed through a grounded theory approach. As a result, I anticipate rich narrative descriptions of the rules and resources that structure stakeholder behavior and the Open Science Framework. By including technology developers, users, and non-users, this research will consider how the variety of stakeholders interact—in conflict and accord—to enact open science.

4.1 Sample and Recruitment

I define three populations: developers of the OSF, users of the OSF, and non-users of the OSF. The sample categories are designed so that recruitment will cover a diversity of perspectives and degrees of engagement with the OSF, prioritizing informants who can discuss recent experiences with the platform. I anticipate conducting 45 interviews—15 each across the three sample populations. However, recruitment will continue until no further insights are emerging from the data, i.e. at the point of data saturation [4].

OSF developers are workers at COS who contribute to the OSF code or who contribute to the planning or execution of the platform's improvement. Thus, programmers and managers who specify requirements are both considered developers. Developers are identified via the COS company directory (cos.io/about/team) or recommendations by other staff members and are sent an invitation to participate via email.

OSF users and non-users are defined in contrast with one another based on their recruitment mechanisms. Users can be researchers (faculty or students) who created an OSF account by uploading a file to OSF Meetings during 2020 or 2021 and who then generated additional activity logs by using the platform further. In contrast, non-users created an OSF account by uploading a file to OSF Meetings during 2020 or 2021 but never performed further actions on osf.io to generate additional activity logs. Users can also be researchers with an OSF account who used the platform in 2020 or 2021, meaning that they generated an activity log during those years. In contrast, non-users were invited in 2020 or 2021 by a co-author who had already made an OSF project, but the non-user never followed through to make the account (generating no activity logs). Finally, non-users can also be matched pairs to an OSF user; these non-users must be in the same department as an interviewed user, must have published in the same venues or used the same manuscript keywords in the last three years, and must express no familiarity with the OSF.

Recruitment of users and non-users is facilitated by COS. Using the operational definitions described above (excluding matched pairs), users and non-users can be programmatically identified. By sending a recruitment email on my behalf, inviting the recruit to sign up to participate, COS protects the privacy of their users. Study participants are given the option to reveal their identities and thus be recognized for their contribution to this research.

I will gain access to observation opportunities by asking COS managers if I may observe team meetings.

4.2 Study Procedures

Semi-structured, hour-long interviews will proceed over Zoom after consent is obtained. I have designed different interview protocols for each of my sample populations. Based on a review of the participant's activity on the OSF and/or Github (where OSF code is stored), I further tailor these protocols to each interviewee. Broadly, developer interviews cover their role at COS, their perception of how users engage with the platform, and their perspective on scientific values and open science. While protocols are specific to the recruitment mechanism, interviews with OSF users and non-users begin with a screensharing tour of their file storage for a research project they describe before turning to sharing practices and OSF familiarity; they close with questions about their perspective on scientific values and open science.

Following interviews, I will manually collect trace data from the OSF and Github. A piece of trace data will include at minimum a time stamp, a participant name or pseudonym, and a description of the action taken, as logged by the system. Aside from serving as data in their own right, trace data can be used as prompts in future interviews [7].

Observation of OSF developer meetings will occur over Zoom. While observing participants, I will take notes digitally. I will refrain from involving myself in discussions, acting as a passive observer rather than involved ethnographer.

Data will be analyzed using a constructivist grounded theory approach [4, 6]. This will involve inductive coding to systematically identify emergent patterns and relationships that can be coalesced into abstract explanations of the phenomenon of interest, the enactment of open science. I will iteratively collect data, conduct initial qualitative coding, conduct focused coding, and, from the emergent codes and categories (sometimes called themes), I will build theory. Coding will be augmented with other analytical strategies like temporal bracketing, visual mapping, and memoing.

5 WORK TO DATE

My dissertation proposal was approved in April 2021. Thus far, I have been determined exempt by the UT Austin IRB, meaning that no signature is required on a consent form. I have conducted six interviews with my most accessible population: OSF developers. I have transcribed and conducted initial coding on these interviews. I have been granted tentative permission to observe meetings of the COS Policy Team and COS Infrastructure Team and have already observed a COS Product Team meeting.

I have collaborated with COS to recruit users and non-users. Invitations have been sent out by COS on a weekly basis. Because of a lack of response, these batches have increased in size over the weeks from 15 invitations to 50 invitations to 100 invitations. The subject line has been revised from its initial version to make clearer that it is not a COS newsletter but an invitation to a study. I have also begun identifying qualifying participants and reaching out via email myself; this has increased response rate. So far, I have conducted and transcribed three interviews with non-users and four interviews with users.

Already, in my interviews I am seeing that (imagined) stakeholders outside of academia are structuring the technical decisions of some OSF developers. These imagined stakeholders are conceived by OSF developers as a public that deserves access to "the best truth[s]" (Yuhuai Liu, developer); they are factored into technical decisions like the data model underlying the platform's search function and the presentation of search results. However, other developers contend that the public and citizen scientists (who are not subject to academic incentive structures) are not in need of tools like the OSF to reduce their reporting bias. Thus, while this latter group concerns themselves with using non-pedantic language to educate scientists about open practices like registration, they abstain from educating non-academics because they are not "under the gun" (David Mellor, developer).

6 CONCLUSION

Open science appears to be the way of the future, enforced and promoted through technology and policy. While this is touted as a boon for the speed and quality of research, it is important to consider voices of dissent and mechanisms for action. By conducting interviews, observation, and trace data analysis, I will develop a rich picture of the OSF as it is approached in practice by a diverse group of open science stakeholders. If open science technologies

are the platforms for modern research, we must also consider how non-users shape and are shaped by them. By including non-users of the OSF in my sample and exploring emergent themes like the importance of the public and citizen science to the platform's development, my work will speak to the boundaries of the community that affects and is affected by open science technology. This study will reveal the tensions negotiated as open science is enacted over digital infrastructure and will enable the thoughtful development of future policies and platforms that embrace the varied concerns of the (re)defined scientific community.

REFERENCES

- [1] 2013. COS: Strategic Plan. <https://docs.google.com/document/u/0/d/17OTxjE5zl34VcXiAORayFOCDIkyRCiefh1ZBN1qEBUc>
- [2] Anne Beaulieu, Andrea Scharnhorst, and Paul Wouters. 2007. Not Another Case Study: A Middle-Range Interrogation of Ethnographic Case Studies in the Exploration of E-Science. *Science, Technology, & Human Values* 32, 6 (2007), 672–692. <http://www.jstor.org/stable/29734013>
- [3] Kirsten Boehner and Carl DiSalvo. 2016. Data, Design and Civics: An Exploratory Study of Civic Tech. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. ACM, San Jose California USA, 2970–2981. <https://doi.org/10.1145/2858036.2858326>
- [4] Kathy Charmaz. 2014. *Constructing Grounded Theory* (2 ed.). SAGE.
- [5] Anthony Giddens. 1984. *The Constitution of Society: Outline of the Theory of Structuration*. University of California Press.
- [6] Barney G. Glaser and Anselm L. Strauss. 1967. *The discovery of grounded theory: strategies for qualitative research*. Aldine Pub. Co., Chicago. <http://hdl.handle.net/2027/hvd.32044018542340>
- [7] Sean Goggins, Christopher Mascaro, and Giuseppe Valetto. 2013. Group informatics: A methodological approach and ontology for sociotechnical group research. *Journal of the American Society for Information Science and Technology* 64, 3 (March 2013), 516–539. <https://doi.org/10.1002/asi.22802>
- [8] Kyle Harp-Rushing. 2020. *Fixing Science: Innovation, Disruption, Maintenance, and Repair at a North American Open Science Non-Profit*. Ph.D. Dissertation. UC Riverside. <https://escholarship.org/uc/item/8rs6279m>
- [9] James Howison and James D. Herbsleb. 2013. Incentives and integration in scientific software production. In *Proceedings of the 2013 conference on Computer supported cooperative work - CSCW '13*. ACM Press, San Antonio, Texas, USA, 459. <https://doi.org/10.1145/2441776.2441828>
- [10] Thomas Parke Hughes. 1976. The Science-Technology Interaction: The Case of High-Voltage Power Transmission Systems. *Technology and Culture* 17, 4 (Oct. 1976), 646. <https://doi.org/10.2307/3103672>
- [11] Nadine Levin and Sabina Leonelli. 2017. How Does One “Open” Science? Questions of Value in Biological Research. *Science, Technology, & Human Values* 42, 2 (March 2017), 280–305. <https://doi.org/10.1177/0162243916672071>
- [12] Donald A. MacKenzie and Judy Wajcman (Eds.). 1999. *The social shaping of technology* (2nd ed ed.). Open University Press, Buckingham [Eng.] ; Philadelphia.
- [13] Eric T Meyer, Ralph Schroeder, and William H Dutton. 2008. The Role of e-Infrastructures in the Transformation of Research Practices and Outcomes. (2008), 8.
- [14] Michael J. Mulkay. 1976. Norms and ideology in science. *Social Science Information* 15, 4-5 (Aug. 1976), 637–656. <https://doi.org/10.1177/053901847601500406>
- [15] National Academies of Sciences, Engineering, and Medicine. 2018. *Open Science by Design: Realizing a Vision for 21st Century Research*. The National Academies Press, Washington, D.C. <https://doi.org/10.17226/25116>
- [16] Brian A. Nosek, J. R. Spies, and M. Motyl. 2012. Scientific Utopia: II. Restructuring Incentives and Practices to Promote Truth Over Publishability. *Perspectives on Psychological Science* 7, 6 (Nov. 2012), 615–631. <https://doi.org/10.1177/1745691612459058>
- [17] Wanda J. Orlikowski. 2000. Using Technology and Constituting Structures: A Practice Lens for Studying Technology in Organizations. *Organization Science* 11, 4 (2000), 404–428. <http://www.jstor.org/stable/2640412>
- [18] Astrid Recker, Stefan Müller, Jessica Trixa, and Natascha Schumann. 2015. Paving the Way For Data-Centric, Open Science: An Example From the Social Sciences. *Journal of Librarianship and Scholarly Communication* 3, 2 (Sept. 2015), eP1227. <https://doi.org/10.7710/2162-3309.1227>
- [19] Jeffrey Robert Spies. 2013. *The Open Science Framework*. Ph.D. Dissertation. The University of Virginia. <https://osf.io/5ezrw>
- [20] James Stewart, Rob Procter, Robin Williams, and Meik Poschen. 2013. The role of academic publishers in shaping the development of Web 2.0 services for scholarly communication. *New Media & Society* 15, 3 (May 2013), 413–432. <https://doi.org/10.1177/1461444812465141>